

Modular arithmetic

MINNESOTA ARML

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A modular arithmetic problem set for all levels. Contributed by Michael Luo.

Algebra mains are better than geo mains by the
AM–GM inequality.

Ali at MOP 2025

1 Yap session (short)

The yap session is intentionally short, since this is a very broad topic. Some problems in [Section 3.3](#) require the notion of a primitive root. If enough people are interested, I'll go over it after my short yap session and everyone else is working on the problem set. Otherwise, you can read [Section 2](#).

Theorem 1.1 (Inverses exist). *Let n be a positive integer and let a be an integer with $\gcd(a, n) = 1$. Then there is a unique b with $0 \leq b < n$ such that $ab \equiv 1 \pmod{n}$. We call b the modular inverse of a modulo n , and b is usually written a^{-1} or $\frac{1}{a}$.*

Essentially, this means that we can do “division” modulo n if the thing we’re dividing by is relatively prime to n . This acts exactly the same as “regular” division. That is, we have

$$\frac{a}{b} \cdot \frac{c}{d} \equiv \frac{ac}{bd} \quad \text{and} \quad \frac{a}{b} + \frac{c}{d} \equiv \frac{ad + bc}{bd}$$

modulo n as long as b and d are relatively prime with n .

Example 1.2. Let n be a positive integer relatively prime to 5, and let x be the multiplicative inverse of 5 modulo n . Show that either the remainder of $2x$ when it is divided by n or the remainder when $3x$ is divided by n is greater than $\frac{n}{2}$.

Theorem 1.3 (Euler’s theorem). *Let n be a positive integer and let a be an integer with $\gcd(a, n) = 1$. Then*

$$a^{\varphi(n)} \equiv 1 \pmod{n}.$$

The case $n = p$ is Fermat’s little theorem, $a^{p-1} \equiv 1 \pmod{p}$.

Theorem 1.4 (Chinese remainder theorem). *Let m and n be relatively prime positive integers. Let a be a positive integer. Then “knowing a modulo mn , and knowing a modulo m and a modulo n are the same thing.” That is, to compute a modulo mn , you only need to compute a modulo m and a modulo n .¹*

¹For abstract algebra enthusiasts, the map $\mathbb{Z}/mn\mathbb{Z} \rightarrow \mathbb{Z}/m\mathbb{Z} \times \mathbb{Z}/n\mathbb{Z}$ given by $x \mapsto (x \bmod m, x \bmod n)$ is a ring isomorphism.

This theorem is extremely important. If you see a problem that asks you to find something modulo $2^3 \cdot 3^2 \cdot 5$, the Chinese remainder theorem tells you it's only necessary to compute it modulo 2^3 , 3^2 , and 5 individually.

Example 1.5 (CMIMC 2023/A3). (★) Compute

$$2022^{2022^{\cdot^{2022}}} \pmod{111}$$

where there are 2022 2022s.

2 Theory problems (boring)

If you're new, skip these and go straight to [Section 3.1](#).

Exercise 2.1. Verify [Theorem 1.1](#) for $n = 15$ by writing down every positive integer less than 15 relatively prime to 15 and its inverse modulo 15.

Exercise 2.2. Let S_n be the set of numbers at most n relatively prime to n . Write down each element of S_n and multiply it by a , then take modulo n . Show that this process gives you S_n back (possibly in a different order). Conclude [Theorem 1.1](#). Hint: first try the case $n = 15$.

Exercise 2.3. From the last exercise, we have the set equivalence $S_n = aS_n$. Take the product of the elements of both sides modulo n , and divide by some number relatively prime to n . Conclude [Theorem 1.3](#).

Let p be a prime. The *order* of an element a modulo p , denoted $\text{ord}_p(a)$, is the smallest m such that $a^m \equiv 1 \pmod{p}$.

Exercise 2.4 (Fundamental theorem of orders). Show that if $a^n \equiv 1 \pmod{p}$, then $\text{ord}_p(a) \mid n$.

A primitive root g modulo p is an element modulo p with order $p - 1$.

Exercise 2.5 (Primitive roots generate everything). For all primitive roots g modulo p ,

$$\{1, 2, \dots, p-1\} = \{g^0, g^1, \dots, g^{p-2}\}.$$

There are primitive roots modulo every prime. In fact, there are primitive roots modulo every power of an odd prime p^α : this means that there exists g such that the numbers modulo p^α relatively prime to p are all hit by some power of g . For example, when $p^\alpha = 3^2$, we have the primitive root 2, since

$$\{1, 2, 4, 5, 7, 8\} = \{1, 2, 4, 8, 7, 5\} \equiv \{2^0, 2^1, 2^2, 2^3, 2^4, 2^5\} \pmod{9}.$$

A useful trick when orders/primitive roots are involved is to take a “discrete logarithm” with the base as a primitive root (as in, represent everything as g^k where g is a primitive root). This turns multiplication into addition, which is much easier to think about.

3 Exercises

Please don't be afraid to ask for help: these problems are designed to be hard (so you can ask me questions)! Problems marked with a (★) are particularly instructive, so if you don't know where to start, start with those. The following are in (approximately) increasing order of difficulty.

3.1 Level 1

Exercise 3.1 (Bustimathon cheese). Write down integers $s_1, s_2, \dots, s_{12} \geq 10^{12}$ such that

$$10 + 2(s_1 + s_2 + \dots + s_{12}) \equiv 2 \pmod{2024}.$$

Solving this problem earned me \$25 on the ARML bus two years ago; ask me about the story.

Exercise 3.2. Compute modulo 4: $17 \cdot 18$, $523 \cdot 421$, 15^{15} , $121 \cdot 122 \cdot 123$, $100!$, $1 \cdot 3 \cdot 5 \cdot 7 \cdot 9$.

Exercise 3.3. Compute $514 \cdot 891 \pmod{11}$.

Exercise 3.4. (★) Compute $98^3 \pmod{100}$ in your head. (You can write stuff on paper to think, but the “good” way of doing this problem should involve no actual multiplication on paper.)

Exercise 3.5. Find $1 + 2 + 3 + \dots + 99$ modulo 100.

Exercise 3.6. Compute 3^{1000} modulo 23.

Exercise 3.7. (★) Compute $24^{50} - 15^{50} \pmod{13}$ in your head. (You can write stuff on paper to think, but the “good” way of doing this problem should involve no actual multiplication on paper.)

Exercise 3.8 (ARML 2018/Super Relay 3). (★) For each integer n , let $f(n)$ be the remainder when $n^2 - 1$ is divided by 8. Compute $f(2024) + f(2025) + f(2026) + f(2027)$.

Exercise 3.9. Compute the inverses of the numbers $1, 2, \dots, 10$ modulo 11. Note that $(a^{-1})^{-1} \equiv a \pmod{11}$, as expected.

3.2 Level 10

Exercise 3.10. Prove that a number is congruent to the sum of its digits modulo 9. Conclude the divisibility test for 9.

Exercise 3.11. Compute $1^2 + 2^2 + \dots + 100^2 \pmod{57}$.

Exercise 3.12. (★) Compute $71^8 \pmod{100}$ in your head. (You can write stuff on paper to think, but the “good” way of doing this problem should involve no actual multiplication on paper.)

Exercise 3.13 (ARML 2018/Tiebreaker 1). The increasing infinite arithmetic sequence of integers $x_1, x_2, x_3, \dots$ contains the terms $17!$ and $18!$. Compute the greatest integer X such that $X!$ must also appear in the sequence.

Exercise 3.14 (Prize 2022/4). (★) Determine the largest integer n such that $n < 103$ and $n^3 - 1 \equiv 0 \pmod{103}$.

Exercise 3.15 (φ is multiplicative). Let φ be the Euler totient function, so $\varphi(n)$ is the number of positive integers at most n and relatively prime to n . Let m and n be relatively prime positive integers. Prove that $\varphi(mn) = \varphi(m)\varphi(n)$.

Exercise 3.16 (Mathleague Invitational 2024/14). (★) Compute $\lfloor \frac{10^{2024}}{10^{88}-7} \rfloor$ modulo 100.

3.3 Level 100

Exercise 3.17 (AIME I 2019/14). Find the least odd prime factor of $2019^8 + 1$.

Exercise 3.18 (AIME I 2024/13). Let p be the least prime number for which there exists an integer n such that $n^4 + 1$ is divisible by p^2 .

- Find p .
- Find the least positive integer m such that $m^4 + 1$ is divisible by p^2 .

Exercise 3.19 (HMMT 2019/Guts 28). (★) The last one is the actual problem, the first two are warm-ups.

- For how many integers a with $1 \leq a \leq 41$ is the sequence $a^1, a^2, a^4, a^8, \dots$ eventually constant modulo 41?
- For how many integers a with $1 \leq a \leq 25$ is the sequence $a^1, a^2, a^4, a^8, \dots$ eventually constant modulo 25?
- For how many integers a with $1 \leq a \leq 100$ is the sequence $a^1, a^2, a^4, a^8, \dots$ eventually constant modulo 100?

Exercise 3.20 (HMMT 2015/Guts 31). (★) Define a *power cycle* to be a set S consisting of the nonnegative integer powers of an integer a , i.e., $S = \{1, a, a^2, \dots\}$ for some integer a . Compute the minimum number of power cycles required such that given any odd integer n , there exists some integer k in one of the power cycles such that $n \equiv k \pmod{1024}$.

Exercise 3.21 (Prize 2022/17). (★) Let O be the set of odd numbers between 0 and 100. Let T be the set of subsets of O of size 25. For any finite subset of integers S , let $P(S)$ be the product of the elements of S . Compute $\sum_{S \in T} P(S) \pmod{17}$.

Exercise 3.22 (OTIS Mock AIME 2024/13). Compute the sum of all integers n such that $1 \leq n \leq 2024$ and exactly one of $n^{22} - 1$ and $n^{40} - 1$ is divisible by 2024.

Exercise 3.23 (PUMaC A 2016/N7). For how many integers n with $2017 \leq n \leq 2017^2$ do we have $n^n \equiv 1 \pmod{2017}$?

Exercise 3.24 (HMMT 2015/Guts 31, but harder). [Exercise 3.20](#) except n does not have to be odd.

3.4 Level 1000

Exercise 3.25 (IMO 2005/4). Determine all positive integers relatively prime to all the terms of the infinite sequence

$$a_n = 2^n + 3^n + 6^n - 1, \quad n \geq 1.$$

Exercise 3.26 (IMO 2024/2). Determine all pairs (a, b) of positive integers for which the sequence

$$a_n := \gcd(a^n + b, b^n + a)$$

is eventually constant.